

```
public static int task4CM(int[] input) {
    HashSet<Integer> numSet = new HashSet<>();
    int maxLength = 0;
    for (int num : input) {
        numSet.add(num);
    }
    for (int num : input) {
        if (!numSet.contains(num - 1)) {
            int currentNum = num;
            int currentLength = 1;
            while (numSet.contains(currentNum + 1)) {
                currentNum++;
                currentLength++;
            }
            maxLength = Math.max(maxLength, currentLength);
        }
    }
    return maxLength;
}

public static void main(String[] args) {
    int[] input = {100, 4, 200, 1, 3, 2, 5};
    int result = task4CM(input);
    System.out.println(result);
}
```